Multi Sensor Logic Gate Elephant Instrusion System

Digital Logic and Computer Organization (DLCO)
Academic Hardware Project

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Abstract

Human–elephant conflict motivates the exploration of practical warning systems for communities near wildlife habitats. This academic project developed a hardware-based, multi-sensor intrusion warning prototype using PIR, pressure and light-dependent resistor (LDR) inputs, digital logic gates and a buzzer. Sensor signals were combined by a logic circuit to activate an audible alarm under the configured input conditions. The group assembled and successfully demonstrated the working prototype to the module lecturer, who provided positive feedback. No camera, microcontroller, IoT connectivity, SMS notification or mobile application was implemented. The prototype demonstrates digital logic and sensor integration; it was not validated as an elephant-specific identification system.

Acknowledgment

We thank our module lecturer, Mr. Thisara Ganegoda, for guidance and feedback, NIBM for supporting the learning environment, and our group members for their contributions to the hardware project.

Introduction

Human–elephant conflict affects agricultural communities in parts of Sri Lanka. Early warning concepts can help motivate safer approaches to monitoring potential intrusions. For the DLCO module, our group designed and constructed a small-scale hardware demonstration of a multi-input warning circuit. Its educational goal was to apply Boolean logic, logic gates and physical sensor signals to an audible alert. The prototype is not a tested wildlife-management product and does not determine animal species.

Objectives

- Build a functioning sensor-based warning circuit using basic digital logic gates.
- Combine multiple input conditions into a buzzer-control output.
- Demonstrate and document the hardware operation and circuit design.
- Explore the limitations of a simple sensor-based warning system.

Methodology and Implementation

Components

The documented components include a PIR sensor, pressure sensor, LDR, dot board, battery/power supply, logic-gate circuitry and buzzer. The original circuit diagram depicts integrated circuits and supporting connections. Component part numbers and precise operating specifications should be checked against the physical build before being stated as verified facts.

Working Principle

The PIR, pressure and LDR inputs provide signals to the digital logic circuitry. AND and OR gate combinations evaluate the configured input states. When the circuit output satisfies the alarm condition, the buzzer sounds. The LDR contributes a light-dependent input. The precise Boolean expression should be read from the original logic diagram and checked against the original truth table before further technical claims are made.

Tools and Scope

The original report lists Tinkercad as a software tool. The implemented demonstration was a physical hardware circuit; it did not use a microcontroller, camera, network connection or mobile alert service. Tinkercad should be retained as a design/simulation tool only if it was actually used by the group.

Initial Design Records

From the original report, the following pictures have been kept. The truth-table values and correspondence to the physical circuit have not been independently revalidated in this portfolio revision, despite the fact that they document the original design.

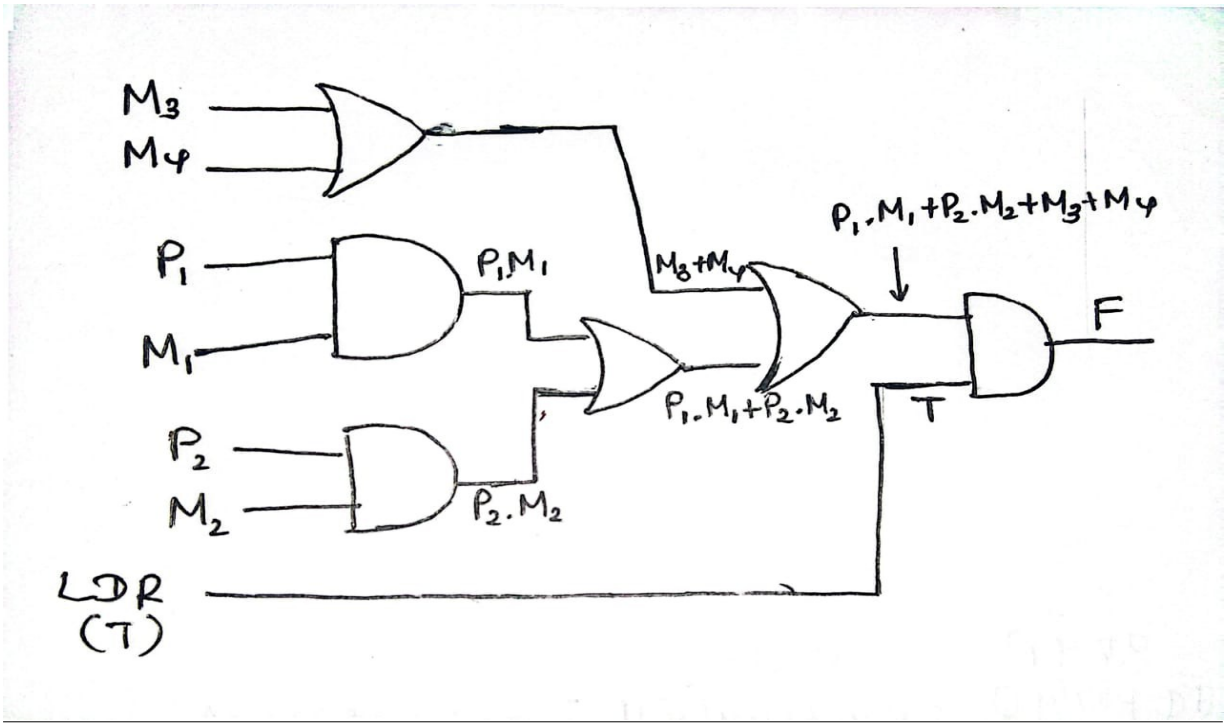
Truth table

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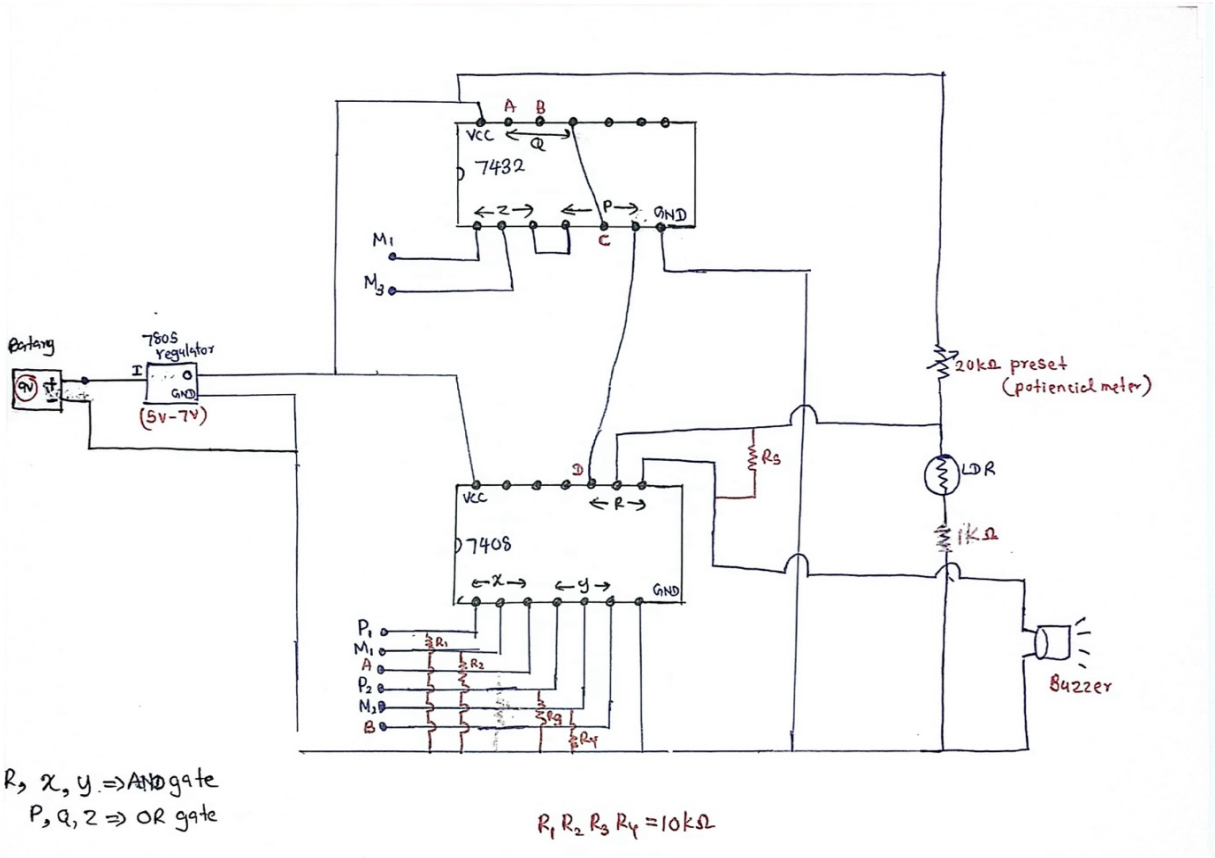
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Logic Gate Diagram



Circuit Diagram



R, X, Y ⇒ AND gate
P, Q, Z ⇒ OR gate

$R_1 R_2 R_3 R_4 = 10k\Omega$

Testing and Results

The team successfully demonstrated that the buzzer would sound under the desired sensor/input conditions after assembling the physical prototype. After watching the demonstration, the module lecturer provided encouraging comments. No quantitative performance claims are made because no recorded measurements were given for detection distance, false-positive rate, long-term reliability, environmental performance, or elephant-specific detection accuracy.

Discussion and Limitations

The prototype demonstrates how a basic audible alarm can be controlled without programming or internet connectivity by combining multiple sensor signals with digital logic. However, an elephant cannot be accurately distinguished from other animals or environmental activity using PIR, pressure, and light-dependent inputs alone. Rather than being a field trial, the demonstration was carried out as academic hardware testing. Unverified are detection range, weather resistance, power endurance, and false-alarm behavior. IoT integration, camera processing, SMS messaging, and remote monitoring were all absent.

Possible Future Improvements

- To identify different animal species, look into additional sensing and proven classification techniques.
- In controlled environments, assess field performance, false alarms, weather resistance, and detection range.
- As a separate future extension, investigate IoT connectivity or remote notifications.
- Before thinking about practical application, enhance enclosure, power management, and safe deployment design.

Conclution

As an academic hardware prototype, the Multi-Sensor Logic Gate Elephant Intrusion Detection System was successfully constructed and demonstrated. When the predetermined conditions were satisfied, it activated a buzzer using sensor inputs and digital logic circuitry. The work gave me hands-on experience with hardware troubleshooting, sensor integration, Boolean logic, and circuit assembly. The prototype is not a verified elephant-recognition or deployed wildlife alert system; rather, it is a proof of concept for a sensor-based intrusion warning circuit.
Portfolio note: The original group report has been revised. It keeps the original design figures while eliminating unsupported implementation and performance claims.